Chapter 3. Elementary Matrix Operations and Systems of Linear Equations

Row operations, rank and Gaussian elimination

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3.1 Elementary matrix operations and elementary matrices

Motivation. Three simple row (or column) operations underlie Gaussian elimination. Each is multiplication by an invertible elementary matrix, so they can be analysed with the matrix algebra of Chapter 2.

Definition (Elementary operations)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, an **elementary row [column] operation** is (1) interchanging two rows [columns]; (2) multiplying a row [column] by a nonzero scalar; (3) adding a scalar multiple of a row [column] to another row [column]. These are of **type** 1, 2, 3.

Example 1

For $A = \left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & -1 & 3 \\ 4 & 0 & 1 & 2 \end{smallmatrix}\right)$: interchanging rows $1, 2$ gives $B = \left(\begin{smallmatrix} 2 & 1 & -1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 0 & 1 & 2 \end{smallmatrix}\right)$; multiplying column 2 by 3 gives C; adding $4\times$row 3 to row 1 gives $M = \left(\begin{smallmatrix} 17 & 2 & 7 & 12 \\ 2 & 1 & -1 & 3 \\ 4 & 0 & 1 & 2 \end{smallmatrix}\right)$.

Definition (Elementary matrix)

An $n \times n$ **elementary matrix** is obtained from I_n by one elementary operation; its type is that of the operation.

Theorem 3.1

If B is obtained from $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ by an elementary row [column] operation, then $B = EA$ $[B = AE]$, where E is the $m \times m$ $[n \times n]$ elementary matrix obtained from I_m $[I_n]$ by the same operation. Conversely, if E is elementary, EA $[AE]$ is obtained from A by the corresponding row [column] operation.

Proof [Supplement]. By Theorem 2.13 (in its row form), the i-th row of EA is (row i of E)$A = \sum_k E_{ik}$(row k of A). Row i of E is e_i^t unless the operation changed it. For type 1 (swap rows p, q), row p of E is e_q^t, so row p of EA is row q of A, and vice versa. For type 2 (row p times c), row p of EA is c times row p of A. For type 3 (add c·row q to row p), row p of E is $e_p^t + ce_q^t$, so row p of EA is row p plus c times row q of A. All other rows are unchanged. Column operations follow by transposing: $(AE)^t = E^tA^t$, and E^t is the elementary matrix of the transposed operation. ■

Example 2

In Example 1, $B = EA$ with $E = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, and $C = AE'$ with $E' = \mathrm{diag}(1, 3, 1, 1)$.

Theorem 3.2

Elementary matrices are invertible, and the inverse of an elementary matrix is elementary of the same type.

Proof. An elementary E comes from I_n by an operation, which can be undone by an operation $\bar{E}$ of the same type: swap the same rows again; multiply by c^{-1}; add $-c$ times the row. By Theorem 3.1, $\bar{E}E = I_n$, so E is invertible with $E^{-1} = \bar{E}$ (a square matrix with a one-sided inverse is invertible, since $\mathsf{L}_{\bar{E}}\mathsf{L}_E = \mathsf{I}$ forces L_E to be one-to-one, hence invertible by Theorem 2.5). ■

3.2 The rank of a matrix and matrix inverses

Motivation. The rank of a matrix is the dimension of its column space. Elementary operations preserve it, reduce every matrix to $\left(\begin{smallmatrix} I_r & O \\ O & O \end{smallmatrix}\right)$, and show that row rank equals column rank. The same operations, applied to $(A \mid I)$, compute inverses.

Definition (Rank of a matrix)

$\operatorname{rank}(A) = \operatorname{rank}(\mathsf{L}_A)$. Thus an $n \times n$ matrix is invertible iff its rank is n (Theorem 2.5 and Corollary 2 of Theorem 2.18).

Theorem 3.3

For linear $\mathsf{T} : V \to W$ between finite-dimensional spaces with ordered bases β, γ: $\operatorname{rank}(\mathsf{T}) = \operatorname{rank}([\mathsf{T}]_\beta^\gamma)$.

Proof [Supplement]. With $A = [\mathsf{T}]_\beta^\gamma$, $\mathsf{L}_A \phi_\beta = \phi_\gamma \mathsf{T}$ (Theorem 2.14). As ϕ_β is onto, $\mathrm{R}(\mathsf{L}_A) = \mathrm{R}(\mathsf{L}_A \phi_\beta) = \phi_\gamma(\mathrm{R}(\mathsf{T}))$, and the isomorphism ϕ_γ preserves dimension. ■

Theorem 3.4

If P, Q are invertible $m \times m$ and $n \times n$, then (a) $\operatorname{rank}(AQ) = \operatorname{rank}(A)$,
(b) $\operatorname{rank}(PA) = \operatorname{rank}(A)$,
(c) $\operatorname{rank}(PAQ) = \operatorname{rank}(A)$.

Proof.

(a) L_Q is onto, so $\mathrm{R}(\mathsf{L}_{AQ}) = \mathsf{L}_A \mathsf{L}_Q(\mathbb{F}^n) = \mathsf{L}_A(\mathbb{F}^n) = \mathrm{R}(\mathsf{L}_A)$.

(b) [Supplement] $\mathrm{R}(\mathsf{L}_{PA}) = \mathsf{L}_P(\mathrm{R}(\mathsf{L}_A))$, and the isomorphism L_P preserves dimension.

(c) Combine (a) and (b). ■

Corollary

Elementary row and column operations preserve rank.

Proof. Theorems 3.1, 3.2, 3.4. ■

Theorem 3.5

The rank of a matrix is the maximum number of linearly independent columns, i.e. the dimension of the span of its columns.

Proof. $\mathrm{R}(\mathsf{L}_A) = \operatorname{span}\{\mathsf{L}_A(e_j)\} = \operatorname{span}\{a_1, \ldots, a_n\}$ by Theorems 2.2 and 2.13(b), where $a_j = Ae_j$ is the j-th column. ■

Example 1

$A = \left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{smallmatrix}\right)$: the first two columns are independent and the third is their sum, so $\operatorname{rank} A = 2$.

Example 2

For $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 1 & 1 & 2 \end{smallmatrix}\right)$, subtract row 1 from rows $2, 3$ to get $\left(\begin{smallmatrix} 1 & 2 & 1 \\ 0 & -2 & 2 \\ 0 & -1 & 1 \end{smallmatrix}\right)$, then subtract multiples of column 1 to clear row 1: the result has two independent columns, so $\operatorname{rank} A = 2$.

Theorem 3.6

If $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ has rank r, then $r \le m$, $r \le n$, and finitely many elementary row and column operations transform A into

$$D = \begin{pmatrix} I_r & O_1 \\ O_2 & O_3 \end{pmatrix} \qquad (O_i \text{ zero matrices}).$$

Proof. If $A = O$, then $r = 0$ and $D = A$. Otherwise we show by induction on m that A can be transformed into D for some r; by Corollary , $r = \operatorname{rank} D = \operatorname{rank} A$, and clearly $r \le m, n$.

Step 1 (a pivot). Some entry $A_{ij} \neq 0$; by row and column interchanges move it to position $(1, 1)$, and scale row 1 to make it 1.

Step 2 (clear). Subtract multiples of row 1 from the other rows and multiples of column 1 from the other columns, getting $\left(\begin{smallmatrix} 1 & 0 \\ 0 & B' \end{smallmatrix}\right)$ with B' of size $(m-1) \times (n-1)$.

Step 3 (recurse). If $m = 1$, we are done. Otherwise, by induction (or trivially if $B' = O$), operations on the last $m-1$ rows and $n-1$ columns bring B' to $\left(\begin{smallmatrix} I_{r-1} & O \\ O & O \end{smallmatrix}\right)$ without disturbing the first row and column. ∎

Example 3

$A = \left(\begin{smallmatrix} 0 & 2 & 4 & 2 & 2 \\ 4 & 4 & 4 & 8 & 0 \\ 8 & 2 & 0 & 10 & 2 \\ 6 & 3 & 2 & 9 & 1 \end{smallmatrix}\right)$: interchanging rows $1, 2$, scaling, and clearing as in the proof leads to $D = \left(\begin{smallmatrix} I_3 & O \\ O & O \end{smallmatrix}\right)$, so $\operatorname{rank} A = 3$.

Corollary 1

If $\operatorname{rank} A = r$, there are invertible B $(m \times m)$ and C $(n \times n)$ with $D = BAC$.

Proof. By Theorem 3.6, $D = E_p \cdots E_1 A G_1 \cdots G_q$ with elementary E_i, G_j (Theorem 3.1); take $B = E_p \cdots E_1$ and $C = G_1 \cdots G_q$, invertible by Theorem 3.2. ∎

Corollary 2

For any $A \in \mathrm{M}_{m\times n}(\mathbb{F})$:
(a) $\operatorname{rank}(A^t) = \operatorname{rank}(A)$;
(b) the rank of A is the maximum number of linearly independent rows, i.e. the dimension of the row space;
(c) the rows and the columns of A span subspaces of the same dimension, equal to $\operatorname{rank} A$.

Proof.

(a) With $D = BAC$, $D^t = C^t A^t B^t$ with B^t, C^t invertible, so by Theorem 3.4, $\operatorname{rank} A^t = \operatorname{rank} D^t = r = \operatorname{rank} A$ (D^t visibly has rank r).

(b) The rows of A are the columns of A^t; apply Theorem 3.5 and (a).

(c) Theorem 3.5 and (b). ∎

Corollary 3

Every invertible matrix is a product of elementary matrices.

Proof. If A is $n \times n$ invertible, $\operatorname{rank} A = n$ and $D = I_n$, so $I_n = E_p \cdots E_1 A G_1 \cdots G_q$ and $A = E_1^{-1} \cdots E_p^{-1} G_q^{-1} \cdots G_1^{-1}$, a product of elementary matrices by Theorem 3.2. ■

Theorem 3.7

For linear $\mathsf{T} : V \to W$, $\mathsf{U} : W \to Z$ (finite-dimensional) and matrices with AB defined:
(a) $\operatorname{rank}(\mathsf{UT}) \le \operatorname{rank}(\mathsf{U})$;
(b) $\operatorname{rank}(\mathsf{UT}) \le \operatorname{rank}(\mathsf{T})$;
(c) $\operatorname{rank}(AB) \le \operatorname{rank}(A)$;
(d) $\operatorname{rank}(AB) \le \operatorname{rank}(B)$.

Proof.

(a) $\mathrm{R}(\mathsf{UT}) = \mathsf{U}(\mathsf{T}(V)) \subseteq \mathsf{U}(W) = \mathrm{R}(\mathsf{U})$.

(c) Apply (a) to $\mathsf{L}_A \mathsf{L}_B = \mathsf{L}_{AB}$.

(d) $\operatorname{rank}(AB) = \operatorname{rank}(B^t A^t) \le \operatorname{rank}(B^t) = \operatorname{rank}(B)$ by (c) and Corollary 2.

(b) With bases α, β, γ, $A = [\mathsf{U}]_\beta^\gamma$, $B = [\mathsf{T}]_\alpha^\beta$: $\operatorname{rank}(\mathsf{UT}) = \operatorname{rank}(AB) \le \operatorname{rank}(B) = \operatorname{rank}(\mathsf{T})$ by Theorems 3.3, 2.11 and (d). ■

Example 4

To find a rank, simplify by operations until independence is visible.
(a) $\left(\begin{smallmatrix} 1 & 2 & 1 & 1 \\ 2 & 1 & -1 & 1 \end{smallmatrix}\right)$ has two independent rows: rank 2.
(b) $\left(\begin{smallmatrix} 1 & 3 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 3 & 0 & 0 \end{smallmatrix}\right)$: row 3 = row 1 − row 2, rank 2.
(c) $\left(\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 5 & 4 \end{smallmatrix}\right)$: subtracting gives rows $(1, 2, 1), (0, -3, -3), (0, 3, 3)$, rank 2.

The inverse of a matrix. The **augmented matrix** $(A \mid B)$ of an $m \times n$ A and $m \times p$ B is the $m \times (n + p)$ matrix with the columns of A followed by those of B; $M(A \mid B) = (MA \mid MB)$. If A is invertible, $A^{-1} = E_p \cdots E_1$ (Corollary 3), so the row operations $E_1, \ldots, E_p$ turn $(A \mid I_n)$ into $(I_n \mid A^{-1})$. Conversely, if row operations turn $(A \mid I_n)$ into $(I_n \mid B)$, then $E_p \cdots E_1 A = I$ and $B = E_p \cdots E_1 = A^{-1}$. If instead a row of zeros appears in the left half, $\operatorname{rank} A < n$ and A is not invertible.

Example 5

For $A = \left(\begin{smallmatrix} 0 & 2 & 4 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{smallmatrix}\right)$, row-reducing $(A \mid I_3)$ gives

$$A^{-1} = \begin{pmatrix} \frac{1}{8} & -\frac{5}{8} & \frac{3}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{2} \\ \frac{3}{8} & -\frac{3}{8} & \frac{1}{4} \end{pmatrix}.$$

Example 6

For $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 5 & 4 \end{smallmatrix}\right)$, reducing $(A \mid I_3)$ produces a zero row on the left ($\operatorname{rank} A = 2$), so A is not invertible.

Example 7

$\mathsf{T}(f) = f + f' + f''$ on $\mathsf{P}_2(\mathbb{R})$: in the standard basis $[\mathsf{T}]_\beta = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ with inverse $\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$. So T is invertible (Corollary 1 of Theorem 2.18) and $\mathsf{T}^{-1}(a_0 + a_1x + a_2x^2) = (a_0 - a_1) + (a_1 - 2a_2)x + a_2x^2$.

3.3 Systems of linear equations: theoretical aspects

Motivation. A system $Ax = b$ is a question about L_A: solutions of $Ax = 0$ form the null space, solutions of $Ax = b$ are a translate of it, and solvability means b lies in the column space, i.e. adjoining b does not raise the rank.

A system of m linear equations in n unknowns is $Ax = b$, with **coefficient matrix** $A \in \mathrm{M}_{m\times n}(\mathbb{F})$; a **solution** is $s \in \mathbb{F}^n$ with $As = b$. The system is **consistent** if its solution set is nonempty.

Example 1

(a) $x_1 + x_2 = 3$, $x_1 - x_2 = 1$ has the unique solution $(2, 1)$.
(b) $x_1 + x_2 + x_3 = 1$ (one equation, three unknowns) has infinitely many solutions, e.g. $(1, 0, 0)$ and $(0, 0, 1)$.
(c) $x_1 + x_2 = 0$, $x_1 + x_2 = 1$ has no solution.

Definition (Homogeneous system)

$Ax = b$ is **homogeneous** if $b = 0$; it always has the solution 0.

Theorem 3.8

The solution set K of a homogeneous system $Ax = 0$ (m equations, n unknowns) is $\mathrm{N}(\mathsf{L}_A)$, a subspace of $\mathbb{F}^n$ of dimension $n - \operatorname{rank}(A)$.

Proof. $K = \{s : As = 0\} = \mathrm{N}(\mathsf{L}_A)$, and $\dim \mathrm{N}(\mathsf{L}_A) = n - \operatorname{rank} \mathsf{L}_A$ by Theorem 2.3. ∎

Corollary

If $m < n$, then $Ax = 0$ has a nonzero solution.

Proof. $\operatorname{rank} A \le m < n$, so $\dim K > 0$. ∎

Example 2

(a) $x_1 + 2x_2 + x_3 = 0$, $x_1 - x_2 - x_3 = 0$: $\operatorname{rank} A = 2$, so $\dim K = 1$, and $K = \operatorname{span}\{(1, -2, 3)\}$.
(b) $x_1 - 2x_2 + x_3 = 0$: $\dim K = 2$, with basis $\{(2, 1, 0), (-1, 0, 1)\}$.

Theorem 3.9

Let K be the solution set of $Ax = b$ and K_H that of $Ax = 0$. For any solution s of $Ax = b$, $K = \{s\} + K_H = \{s + k : k \in K_H\}$.

Proof. If $w \in K$, then $A(w - s) = b - b = 0$, so $w = s + (w - s) \in \{s\} + K_H$. Conversely $A(s + k) = b + 0 = b$. ∎

Example 3

(a) $x_1+2x_2+x_3=7$, $x_1-x_2-x_3=-4$ has solution $s=(1,1,4)$, so $K=\{(1,1,4)+t(1,-2,3): t\in\mathbb{F}\}$.
(b) $x_1-2x_2+x_3=4$: $K=\{(4,0,0)+t_1(2,1,0)+t_2(-1,0,1)\}$.

Theorem 3.10

Let $Ax=b$ have n equations and n unknowns. If A is invertible, the unique solution is $A^{-1}b$. Conversely, if the system has exactly one solution, A is invertible.

Proof. If A is invertible, $A(A^{-1}b)=b$, and $As=b$ implies $s=A^{-1}b$. Conversely, if s is the only solution, then by Theorem 3.9 $K_H=\{0\}$, so $\mathrm{N}(\mathsf{L}_A)=\{0\}$ and A is invertible by Theorems 2.5 and 2.18. ■

Example 4

$2x_2+4x_3=2$, $2x_1+4x_2+2x_3=3$, $3x_1+3x_2+x_3=1$ has the invertible coefficient matrix of Example 5 of Section 3.2, so the unique solution is

$$A^{-1}\begin{pmatrix}2\\3\\1\end{pmatrix}=\begin{pmatrix}-\frac{7}{8}\\ \frac{5}{4}\\ -\frac{1}{8}\end{pmatrix}.$$

Theorem 3.11

$Ax=b$ is consistent iff $\operatorname{rank}(A)=\operatorname{rank}(A\mid b)$.

Proof. Consistency means $b\in\mathrm{R}(\mathsf{L}_A)=\operatorname{span}\{a_1,\dots,a_n\}$ (columns), i.e. $\operatorname{span}\{a_1,\dots,a_n\}=\operatorname{span}\{a_1,\dots,a_n,b\}$, i.e. (Theorem 1.11, the first span lying in the second) the two spans have equal dimension: $\operatorname{rank}A=\operatorname{rank}(A\mid b)$ by Theorem 3.5. ■

Example 5

For $x_1+x_2=0$, $x_1+x_2=1$: $\operatorname{rank}A=1$ but $\operatorname{rank}(A\mid b)=2$, so the system is inconsistent.

Example 6

Is $(3,3,2)\in\mathrm{R}(\mathsf{T})$ for $\mathsf{T}(a_1,a_2,a_3)=(a_1+a_2+a_3,\ a_1-a_2+a_3,\ a_1+a_3)$? With $A=\left(\begin{smallmatrix}1&1&1\\1&-1&1\\1&0&1\end{smallmatrix}\right)$, $\operatorname{rank}A=2$ and $\operatorname{rank}(A\mid b)=3$. So no.

3.3.1 An application: Leontief's input–output models

Closed model. Three producers (food, clothing, housing) consume fractions $A_{ij}\ge 0$ of commodity j; each column of A sums to 1. With incomes p, survival requires the **equilibrium condition** $Ap=p$, i.e. $(I-A)p=0$ with $p\ge 0$. (For vectors, $b\ge c$ [$b>c$] means entrywise; b is **nonnegative** [**positive**] if $b\ge 0$ [$b>0$].) The weaker $Ap\le p$ already forces $Ap=p$: otherwise $\sum_i p_i>\sum_i\sum_j A_{ij}p_j=\sum_j p_j$. For

$$A=\begin{pmatrix}0.40&0.20&0.20\\0.10&0.70&0.20\\0.50&0.10&0.60\end{pmatrix},\qquad p=\begin{pmatrix}0.25\\0.35\\0.40\end{pmatrix}$$

solves it: incomes in the ratio $5 : 7 : 8$.

Theorem 3.12

Let A be an $n \times n$ input–output matrix (nonnegative entries, columns summing to 1) of the form $A = \left(\begin{smallmatrix} B & C \\ D & E \end{smallmatrix}\right)$, with D a positive $1 \times (n-1)$ vector and C a positive $(n-1) \times 1$ vector. Then $(I - A)x = 0$ has a one-dimensional solution space generated by a nonnegative vector.

Proof [Supplement]. The book cites the proof. Write $x = (y, x_n)$ with $y \in \mathbb{R}^{n-1}$.

Step 1 (reduction). The columns of $I - A$ sum to 0, so the last equation is minus the sum of the others. Hence $(I - A)x = 0$ iff $(I - B)y = x_n C$.

Step 2 ($I - B$ is invertible). The column sums of B are $1 - D_j < 1$. If $(I - B)y = 0$ with $y \neq 0$, then $|y_i| \leq \sum_j B_{ij}|y_j|$, and summing over i gives $\sum_i |y_i| \leq \sum_j (1 - D_j)|y_j| < \sum_j |y_j|$, a contradiction.

Step 3 (the solution space). By Steps 1–2 the solutions are $x_n((I - B)^{-1}C, 1)$: a one-dimensional space generated by $v = (y_0, 1)$, where $(I - B)y_0 = C$.

Step 4 ($y_0 \geq 0$). Let $N = \{i : (y_0)_i < 0\}$ and suppose $N \neq \emptyset$. For $i \in N$, $(y_0)_i = \sum_j B_{ij}(y_0)_j + C_i > \sum_{j \in N} B_{ij}(y_0)_j$, since the terms with $j \notin N$ and C_i are nonnegative and $C_i > 0$. Hence

$$\sum_{i \in N} |(y_0)_i| < \sum_{j \in N} \Big(\sum_{i \in N} B_{ij} \Big) |(y_0)_j| \leq \sum_{j \in N} |(y_0)_j|,$$

a contradiction. So $v \geq 0$. ∎

Open model. With outside demands $d \geq 0$ and A_{ij} the amount of commodity i needed per unit of commodity j (nonnegative, column sums ≤ 1), we need $x \geq 0$ with $(I - A)x = d$. If $(I - A)^{-1}$ exists and is nonnegative, $x = (I - A)^{-1}d$; this holds when $A^k \to O$, since then $(I - A)^{-1} = I + A + A^2 + \cdots$ (Section 5.3). For example, with

$$A = \begin{pmatrix} 0.30 & 0.20 & 0.30 \\ 0.10 & 0.40 & 0.10 \\ 0.30 & 0.20 & 0.30 \end{pmatrix}, \qquad (I - A)^{-1} = \begin{pmatrix} 2.0 & 1.0 & 1.0 \\ 0.5 & 2.0 & 0.5 \\ 1.0 & 1.0 & 2.0 \end{pmatrix},$$

a demand $d = (30, 20, 10)$ is met by $x = (I - A)^{-1}d = (90, 60, 70)$.

3.4 Systems of linear equations: computational aspects

Motivation. Row operations do not change the solution set. Gaussian elimination brings any augmented matrix to its unique reduced row echelon form, from which the general solution and a basis of the homogeneous solutions can be read off. The same form exposes which columns of a matrix are independent.

Definition (Equivalent systems)

Two systems are **equivalent** if they have the same solution set.

Theorem 3.13

If C is an invertible $m \times m$ matrix, $(CA)x = Cb$ is equivalent to $Ax = b$.

Proof. $As = b \Rightarrow CAs = Cb$, and $CAs = Cb \Rightarrow As = C^{-1}CAs = C^{-1}Cb = b$. ∎

Corollary

If $(A' \mid b')$ comes from $(A \mid b)$ by finitely many elementary row operations, $A'x = b'$ is equivalent to $Ax = b$.

Proof. $(A' \mid b') = C(A \mid b) = (CA \mid Cb)$ with C a product of elementary, hence invertible, matrices. ■

Definition (Reduced row echelon form)

A matrix is in **reduced row echelon form** if
(a) every nonzero row precedes every zero row;
(b) the first nonzero entry of each row is the only nonzero entry in its column;
(c) the first nonzero entry of each row is 1 and lies to the right of the first nonzero entry of the preceding row.

Gaussian elimination. *Forward pass:* in the leftmost nonzero column, bring a nonzero entry to the top row (type 1), scale it to 1 (type 2), and clear the entries below it (type 3). Repeat on the submatrix below and to the right. *Backward pass:* starting from the last nonzero row, clear the entries above each leading 1. For example

$$\begin{cases} 3x_1 + 2x_2 + 3x_3 - 2x_4 = 1 \\ x_1 + x_2 + x_3 = 3 \\ x_1 + 2x_2 + x_3 - x_4 = 2 \end{cases} \quad \rightsquigarrow \quad \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 3 \end{array}\right),$$

so $x_1 = 1 - x_3$, $x_2 = 2$, $x_4 = 3$, with x_3 free.

Example 1

(a) The final matrix above is in reduced row echelon form.
(b) $\left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{smallmatrix}\right)$ is not: the leading 1 of row 1 shares its column with a nonzero entry, violating (b), and the leading 1 of row 3 is not to the right of that of row 2, violating (c).
(c) $\left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$ is in reduced row echelon form; $\left(\begin{smallmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$ is not (leading entry 2).

Theorem 3.14

Gaussian elimination transforms any matrix into reduced row echelon form.

Proof [Supplement]. After the forward pass, each nonzero row has leading entry 1 strictly to the right of the leading entry of the row above, below each leading 1 there are zeros, and zero rows are at the bottom: (a) and (c) hold. The backward pass clears the entries above each leading 1 by adding multiples of its row. This does not change the leading entries or the zeros to their left, since the row used has zeros to the left of its leading 1. So (b) holds as well. ■

Reading off the solution. Let $(A \mid b)$ be in reduced row echelon form with r nonzero rows and no row $(0 \cdots 0 \mid 1)$ (otherwise there is no solution). Solve each equation for its leading variable and set the remaining $n - r$ (free) variables equal to parameters $t_1, \ldots, t_{n-r}$. For

$$\begin{cases} 2x_1 + 3x_2 + x_3 + 4x_4 - 9x_5 = 17 \\ x_1 + x_2 + x_3 + x_4 - 3x_5 = 6 \\ x_1 + x_2 + x_3 + 2x_4 - 5x_5 = 8 \\ 2x_1 + 2x_2 + 2x_3 + 3x_4 - 8x_5 = 14 \end{cases} \quad \rightsquigarrow \quad \left(\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 3 \\ 0 & 1 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right),$$

the general solution is

$$x = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + t_1 \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} 2 \\ -1 \\ 0 \\ 2 \\ 1 \end{pmatrix}.$$

Theorem 3.15

Let $Ax = b$ have r nonzero equations in n unknowns, $\operatorname{rank} A = \operatorname{rank}(A \mid b)$, and $(A \mid b)$ in reduced row echelon form. Then (a) $\operatorname{rank} A = r$;
(b) if the procedure gives $s = s_0 + t_1 u_1 + \cdots + t_{n-r} u_{n-r}$, then $\{u_1, \ldots, u_{n-r}\}$ is a basis of the homogeneous solution set K_H and s_0 is a solution.

Proof.

(a) The system is consistent, so no row is $(0 \cdots 0 \mid 1)$. Each of the r rows has a leading 1 in A, in a column that is e_i; these r columns are independent, and the column space lies in $\mathbb{F}^r$ (rows below r are absent). So $\operatorname{rank} A = r$.

(b) Taking all $t_i = 0$ shows $s_0 \in K$. By Theorem 3.9, $K_H = \{-s_0\} + K = \operatorname{span}\{u_1, \ldots, u_{n-r}\}$, and $\dim K_H = n - r$ (Theorem 3.8). A spanning set of $n - r$ vectors in an $(n-r)$-dimensional space is a basis. ■

What the reduced row echelon form says. Let A have columns $a_1, \ldots, a_n$ and reduced row echelon form $B = CA$ (C invertible) with columns $b_j = Ca_j$. Since L_C is an isomorphism, linear relations among the b_j are exactly those among the a_j.

Theorem 3.16

Let $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ have rank $r > 0$ and reduced row echelon form B. Then
(a) B has exactly r nonzero rows;
(b) for each $i \le r$ there is a column $b_{j_i} = e_i$ ($j_1 < \cdots < j_r$);
(c) the columns $a_{j_1}, \ldots, a_{j_r}$ of A are linearly independent;
(d) if column k of B is $d_1 e_1 + \cdots + d_r e_r$, then $a_k = d_1 a_{j_1} + \cdots + d_r a_{j_r}$.

Proof.

(a) $\operatorname{rank} B = \operatorname{rank} A = r$ (Corollary to Theorem 3.4), and the nonzero rows of B are independent (each has a 1 where the others have 0), so there are r of them (Corollary 2 of Theorem 3.6).

(b) Let j_i be the column of the leading 1 of row i; by (b), (c) of the definition, $b_{j_i} = e_i$ and $j_1 < \cdots < j_r$.

(c) $b_{j_1}, \ldots, b_{j_r}$ are independent, and $\mathsf{L}_{C^{-1}}$ preserves independence.

(d) Every column of B lies in $\operatorname{span}\{e_1, \ldots, e_r\}$ (rows $> r$ are zero), so $b_k = \sum d_i e_i = \sum d_i b_{j_i}$; apply C^{-1}. ■

Corollary

The reduced row echelon form of a matrix is unique.

Proof. By Theorem 3.16, j_1 is the first column of A that is nonzero, and in general j_i is the first column not in the span of the preceding columns. Every other column b_k has coordinates d_i determined by the unique relation $a_k = \sum d_i a_{j_i}$ (a_{j_i} independent). So B is determined by A. ■

Example 2

$A = \begin{pmatrix} 2&4&6&2&4 \\ 1&2&3&1&1 \\ 2&4&8&0&0 \\ 3&6&7&5&9 \end{pmatrix}$ has reduced form $B = \begin{pmatrix} 1&2&0&4&0 \\ 0&0&1&-1&0 \\ 0&0&0&0&1 \\ 0&0&0&0&0 \end{pmatrix}$. So $\operatorname{rank} A = 3$, columns $1, 3, 5$ of A are independent, $a_2 = 2a_1$ and $a_4 = 4a_1 - a_3$.

Example 3

To reduce $S = \{2+x+2x^2+3x^3,\ 4+2x+4x^2+6x^3,\ 6+3x+8x^2+7x^3,\ 2+x+5x^3,\ 4+x+9x^3\}$ to a basis of $\operatorname{span}(S)$: the coordinate vectors are the columns of A in Example 2, so the first, third and fifth polynomials form a basis.

Example 4

$V = \{x \in \mathbb{R}^5 : x_1 + 7x_2 + 5x_3 - 4x_4 + 2x_5 = 0\}$ has dimension 4, with basis $\{(-7,1,0,0,0), (-5,0,1,0,0), (4,0,0,1,0), (-2,0,0,0,1)\}$. To extend the independent set $S = \{(-2,0,0,-1,-1), (1,1,-2,-1,-1), (-5,1,0,1,1)\} \subseteq V$ to a basis, row-reduce the matrix whose columns are the vectors of S followed by this basis. The pivot columns are $1, 2, 3, 6$, so $S \cup \{(4,0,0,1,0)\}$ is a basis of V.